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Energy-Efficient Visual Control of a Self-Balancing Robot using Spiking Neural Networks: A Neuromorphic Approach at the Edge

by

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Abstract

Visual recognition and control in mobile robotics typically relies on computationally intensive Convolutional Neural Networks(CNNs), which poses significant challenges for battery-powered embedded systems, particularly dynamically unstable platforms such as self-balancing robots. Conventional frame-based CNNs process redundant video data regardless of scene dynamics, resulting in substantial energy consumption and processing latency that constrain real-time performance at the edge.

To address these limitations, this dissertation proposes a novel neuromorphic visual control framework based on the Robot Operating System (ROS 2 Humble). The system utilizes Difference Encoding to simulate biological retinal mechanisms for dynamic feature extraction and integrates a custom-designed Spiking Convolutional Neural Network (SCNN). Furthermore, to bridge the gap between algorithmic complexity and embedded constraints, a high-performance C++ inference engine is developed to deploy the SNN within the real-time control loop of the robot.

Experimental results demonstrate that in visual line-following and target tracking tasks, the proposed SNN architecture maintains control accuracy comparable to traditional deep learning approaches while reducing theoretical Synaptic Operations (SOPs) by approximately [X]

Contents

1 Introduction

1.1 Background and Motivation

The proliferation of autonomous mobile systems has revolutionized sectors ranging from industrial logistics to domestic services. At the core of these systems lies visual perception and control, which enables robots to perceive and interpret unstructured environments and execute complex navigation tasks[1]. In recent years, Convolutional Neural Networks(CNNs) have established themselves as the state-of-art solution for robotic vision due to their superior performance in feature extraction and pattern recognition[2].

However, the deployment of Deep Learning models on some mobile robotic platforms (most of them are resource-constrained) introduces a fundamental trade-off between computational performance and energy efficiency. Unlike cloud-based AI systems with almost unlimited computational and energy resources, mobile robots and embedded systems– especially micro-robots and self-balancing robots– operate under significant constraints and challenges due to their limites computational resources, including memory, processing power, and energy consumption[3].

The continuous processing of computionally intensive CNNs on battery-powered embedded processors often leads to rapid depletion and thermal throttling, which severely limites the operational duration of the robot and its ability to perform real-time tasks.

To address these challenges, Neuromorphic Computing has emerged as a promising paradigm shift in embedded Artificial Intelligence. Inspired by the energy-efficient mechanisms of the biological brain, Spiking Neural Networks(SNNs) process information using discrete, sparse events(Spikes) rather than continuous numerical values, can offer order-of-magnitude improments in energy efficiency compared to traditional Artificial Neural Networks(ANNs)[5]. Consequently, investigating the application of SNNs in Edge Robotics represents a critical frontier in modern robotic engineering.

1.2 Problem Statement

Despite their theoretical potential, the pratical deployment of SNNs on physical rototic platforms faces several critical engineering challenges:

Computational Efficiency on Non-Neuromorphic Hardware: While SNNs are designed for specialized neuromorphic chips (e.g., Intel Loihi), most edge robots, such as the Yahboom self-balancing platform, rely on general-purpose microcontrollers like the STM32 and ESP32. Achieving high-performance SNN inference on these platforms requires rigorous optimization of software-hardware co-design.

Sensory Integration and Real-time Control: Autonomous systems such as two-wheeled self-balancing robots are dynamically unstable and require millisecond-level feedback loops. Conventional frame-based vision sensors generate redundant data, which contradicts the event-driven nature of SNNs. Developing a visual encoding mechanism that can bridge frame-based input with spike-based processing isessential for maintaining system stability.

The Sim-to-Real Gap: Many SNN studies remain confined to theoretical simulations or static dataset benchmarks. There is a pressing need for empirical research that demonstrates the feasibility of SNNs in managing closed-loop, real-time control tasks on physical hardware.

1.3 Research Aims and Objectives

The primary aim of this research is to design and implement a high-performance, energy-efficient visual control framework for a self-balancing robot, leveraging the sparse compurting characteristics of SNNs. The research will focus on achieving real-time, colsed-loop navigation by optimizing the synergy between neuromorphic algorithms and embedded hardware. To achieve this aim, the following objectives have been established:

Objective 1: Development of a Hierarchical Control Architecture
To establish a multi-layer control system using ROS 2 Humble. This involves utilizing an ESP32-S3 as a high-speed communication gateway to bridge the high-level SNN cognitive logic(running on a host machine) with the low-level reflexive control (implemented on an STM32F103). The architecture will leverage micro-ROS to ensure synchronized data exchange between the SNN decision-making and the PID balancing loops.

Objective 2: Design of a Retina-inspired Visual Encoder
To develop a visual encoding scheme based on temporal difference algorithms. By computing the pixel-level intensity changes between successive frames, the system converts standaed RGB video streams into sparse, event-based spike tensors. This process aims to minimize redundant computation at the edge, mirroring the biological efficiency of the human retina.

Objective 3: Implementation of an Optimized C++ Inference Engine
To engineer a dedicated SNN inference engine using the Eigen library within a C++ environment. Utilizing software-hardware co-design techniques, this objective focuses on minimizing processing latency to meet the millisecond-level responsiveness required for dynamically unstable platforms.

Objective 4: Performance Evalution and Benchmarking
To quantitatively access the navigation accuracy and energy efficiency of the system. Using Synaptic Operations(SPOs) as the primary metric, the SNN-based controller will be benchmarked against traditional CNNs to validate its energy-saving potential in edge robotics applications.

1.4 Research Significance

This research contributes to the field of Edge Artifical Intelligence(Edge AI) by bridging the gap between biological inspiration and practical robotic engineering:

1. **Practical Framework for Spike-based Robotics**
   By implementing SNNs on a cost-effective, multi-processor platform(integrating STM32 and ESP32), this work provides a scalable bileprint for the transition from theoretical simulation to physical deployment. It validates the robustness of event-driven vision in stabilizing dynamically unstable systems, offering a viable path for the development of long-endurance autonomous embedded devices.

2. **Advancement of Software-Hardware Co-optimization**
   The research demonstrates a rigorous co-optimization straegy, where the high-level C++ SNN engine is harmomized with the low-level micro-ROS communicationlayer. This approach addresses the critical constraints of Edge Intelligence, such as limited memory and power budgets, while ensuring the millisecond-level responsiveness required for real-time control.

3. **Integration of Neuromorphic Theory and Nanostructure Horizons**
   This dissertation establishes a computational foundation for the future migration of AI from general-purpose CPUs to specialized neuromorphic architectures. By validating SNN algorithmic efficiency on standard CMOS-based hardware, the research provides critical performance benchmarks and architectural insights. These findings are essential for the future design of dedicated Neuromorphic Systems-on-Chip (NSoC), facilitating the evolution toward more bio-plausible and energy-efficient digital integrated circuits.

1.5 Thesis Structure

The remainder of this dissertation is organized into the following seven chapters:

- **Chapter 2: Literature Review** examines the theoretical foundations of Spiking Neural Networks (SNNs) alongside classical control theories. It reviews the state-of-the-art in neuromorphic sensing, Real-Time Operating Systems (RTOS) for robotics, and the stability challenges inherent in inverted pendulum systems.

- **Chapter 3: System Methodology** proposes a decoupled hierarchical architecture. It details the mathematical modeling of the self-balancing robot, the design of the cascaded PID control loops, and the development of a "Retina-inspired" visual encoding algorithm for spike-based perception.

- **Chapter 4: Implementation and Engineering** describes the multi-processor hardware integration involving the STM32F103 and ESP32-S3. This chapter focuses on the embedded software engineering aspects, including RTOS task prioritization, micro-ROS middleware configuration, and the development of an optimized C++ SNN inference engine.

- **Chapter 5: Experiments and Results** presents empirical data from physical robot trials. It evaluates the system's performance across multiple dimensions, including balancing stability (pitch jitter), navigation accuracy, and computational energy efficiency quantified by Synaptic Operations (SOPs).
- **Chapter 6: Discussion** provides a critical analysis of the experimental findings, focusing on the trade-offs between biological plausibility and real-time engineering constraints. It also discusses the system's robustness in Edge AI scenarios.
- **Chapter 7: Conclusion** summarizes the key engineering contributions and academic insights of the research, offering recommendations for future advancements in neuromorphic embedded systems and hardware-in-the-loop control.

Set II(a):	Two-Dimensional Look-up Table for Δ,Γ:
	$K = 80, \delta = 0, \sigma = 0.2, r = 0.01 : 0.01 : 0.11, S = 70 : 1 : 80$
Set II(b):	Two-Dimensional Look-up Table for ν:
	$K = 80, r = 0.06, \delta = 0, \sigma = 0.10 : 0.05 : 0.60, S = 70 : 1 : 80$

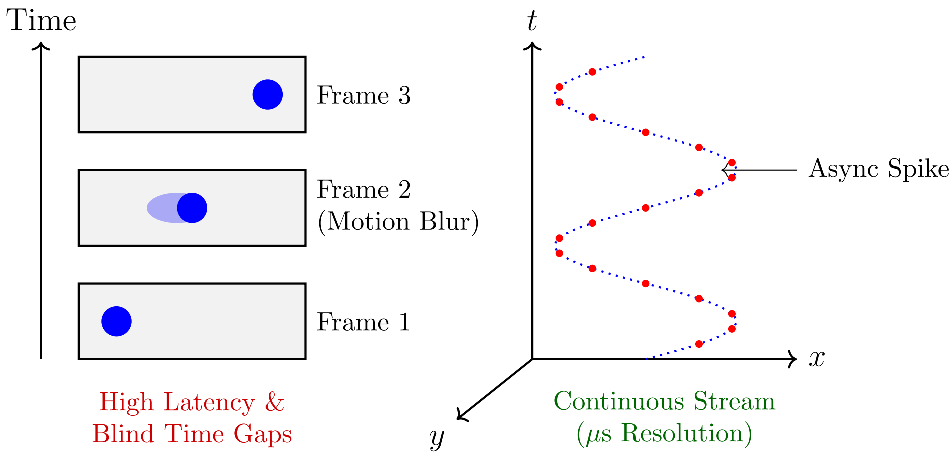
Table 1: Parameters for test

2 Literature Review

2.1 Neuromorphic Computing and Spiking Neural Networks

2.1.1 From Frame-based to Event-based Perception

Conventional Artificial Neural Networks(ANNs), particularly Convolutional Neural Networks(CNNs), rely on dense, frame-synchronous processing, where information is represented by continuous-valued floating-point activations. While effective for static image recognition, this paradigm exhibits inherent computational redundancy: the entire pixel array is re-evaluated via matrix multiplications at each time step, regardless of temporal variations in the scene. For a dynamically unstable system such as a self-balancing robot-modeled as an inverted pendulum-this redundancy imposes a fixed latency floor(bounded by the framerate) and incurs excessive computational overhead, potentially compromising the bandwidth of the stabilization control loop [1].



(a) Frame-based Vision **(b) Event-based Vision**

Figure 1: Visual comparison of acquisition paradigms. **(a)** Standard frame-based cameras capture discrete snapshots with fixed intervals, suffering from blind spots between frames and motion blur. **(b)** Event-based sensors record asynchronous spikes with microsecond resolution, forming a continuous trajectory in space-time ideal for high-speed control.

In contrast, Spiking Neural Networks(SNNs)-conceptualized as the third generation of neural networks—introduce a fundamental paradigm shift towards neuromorphic, event-driven computing. Information is encoded in the precise timing of discrete, binary impulses known as "spikes." This asynchronous processing mechanism confers two critical advantages for embedded robotic control:

1. **Temporal Sparsity and Energy Efficiency:** Neuronal activation is conditional, triggered only when the membrane potential accumulates sufficient stimuli to breach a threshold. Consequently, static background elements in a visual navigation task elicit negligible spiking activity. This sparsity allows the hardware to replace the power-hungry Multiply-Accumulate (MAC) operations typical of conventional ANNs with energy-efficient, event-driven Accumulate (AC) operations [2]. This effectively

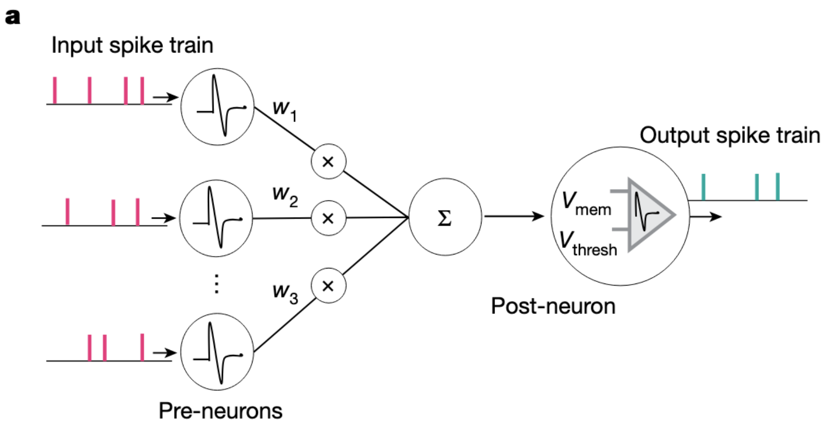


Figure 2: Schematic of pre-synaptic neurons transmitting spike trains to a post-synaptic neuron via synaptic weights. Adapted from [2].

decouples computational cost from input resolution, significantly reducing the dynamic power consumption and processing load on the host CPU.

2. **High Temporal Resolution and Low Latency:** Unlike frame-based systems constrained by the Nyquist sampling rate of standard cameras (e.g., a fixed 33ms latency at 30Hz), SNNs operate on continuous time dynamics. This allows for pseudo-continuous processing of visual inputs, theoretically enabling microsecond-level responsiveness to sudden environmental changes, which is imperative for maintaining stability in high-frequency feedback loops.

2.2 Challenges in Deploying SNNs on Embedded Robotics Platforms

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References

[1] Gallego, G., et al. Event-based vision: A survey. *IEEE TPAMI 44*, 1 (2020), 154–180.

[2] Roy, K., Jaiswal, A., and Panda, P. Towards spike-based machine intelligence with neuromorphic computing. *Nature 575*, 7784 (2019), 607–617.